⚡ STRATEGIC RE-CAP & MANDATE | LIGHTHOUSE CO-DESIGN

Open-Source Foundation Blueprints & Agents: Replicating the Revolut Win

🏆 THE LIGHTHOUSE PROOF POINT: REVOLUT PRAGMA + NVIDIA

- 🌐 **Unprecedented Scale:** Trained Transaction FMs on 24B banking events across 111 countries (NeMo + H100s + cuDF)
- 📈 **Proven Production ROI:** +130% PR-AUC credit risk uplift | +65% fraud detection recall | +41% recommendations
- ⚡ **Architectural Shift:** Single behavioral foundation model replaces dozens of hand-crafted XGBoost pipelines

🌐 1. OPEN-SOURCE STRATEGY & IMPACT

- 🏛️ **Paradigm Shift:** Bespoke ML services ➔ Open foundation blueprints
- 🔓 **100% Extensible:** Any bank adapts to private schemas with zero lock-in
- ⏱️ **Retires 80% Eng Tax:** Eliminates manual SQL joins & static feature store debt
- 🔄 **Multi-Task Core:** Powers Credit, Fraud, AML, and Identity for banks/FIS simultaneously
- 📈 **Fast Delivery:** Shrinks 6-month enterprise POC cycles down to 14 days

🎯 2. GLOBAL BANK REPLICATION ROADMAP

- 🚀 **Revolut:** Expand 1D PRAGMA ➔ Multi-hop Graph FMs for mule rings
- 🇮🇹 **Intesa Sanpaolo:** Behavioral auth against synthetic voice & APP scams
- 🇪🇸 **Santander:** Consolidate custom GNNs ➔ Unified KumoRFM-2
- 🇮🇳 **NPCI (India UPI):** Streaming graphs scoring 15B+ monthly txs (<50ms)
- 🇺🇸 **US Tier-1s:** Zero-data-egress; 100% SR 11-7 audits
- 💎 **Jure Mandate:** 'The less is there, the better' — lean, composable FSI stack

🛡️ PHILOSOPHY & FOUNDATIONAL STRATEGY

Core Operating Tenets for NVIDIA & Kumo.ai in FSI & Risk

🎯 TENET 1: UPSTREAM IDENTITY

- 🔍 **Root Origin:** Fraud is an upstream identity compromise, not downstream payment
- 🛑 **Score Origin:** Payment auth is the crime scene; intercept syndicates early
- 🕸️ **Graph View:** Isolated transaction checks miss multi-hop collusion

🤝 TENET 4: MEET BANKS WHERE THEY ARE

- 🌉 **Pragmatic Bridge:** Meet banks where they are. (with **our agents**)
- 🏢 **Native Systems:** Integrate directly with Snowflake, Databricks, Kafka, & core rails
- ⏱️ **Fast Delivery:** Containerized zero-egress NIMs proving lift in 14-day shadow POCs

⚡ TENET 2: HARD FOUNDATIONAL PRIMITIVES

- 🔬 **Unique Moat:** Build what only NVIDIA + Kumo can solve. Mission: Solving impossible problems + accelerated computing + full-stack AI.
- 🚫 **No Wrappers:** Avoid basic feature stores, dashboards, or generic LLM wrappers
- 🏗️ **Hard Tech:** Own 100B-edge GPU scaling, relational transfer, & sub-30ms serving

💎 TENET 5: THE LESS, THE BETTER

- 📦 **Minimal Stack:** Reject monolithic suites in favor of lean, composable primitives
- 🚀 **Maximum Velocity:** Modular engines drop cleanly into existing bank architectures
- 🎯 **Sharp Focus:** Build foundational primitives that scale horizontally across tiers

🔗 TENET 3: ZERO CRITICAL GAPS

- 🛡️ **Total Coverage:** Zero blind spots across latency, data scale, & graph topologies
- ⚡ **Full Pipeline:** Bridge card auth (<30ms), instant rails (UPI), & warehouse graphs
- 🧩 **Unified Stack:** Prevent banks from stitching together brittle point solutions

🔄 TENET 6: DUAL-LOOP SYMBIOSIS & UPSTREAM AGENTS

- 🔁 **Closed-Loop:** HITL ➔ AIITL for throughput; AIITL ➔ HITL for 100% Quality coverage
- 🤖 **Upstream Agent Fleets:** Coding, Ops, & CX swarms batting with our models.
- 🎥 **Multi-Modal Fusion:** Ingests video (IDV/liveness), images (cards/docs), tabular & graphs
- ⚡ **Dynamic Routing:** Cuts friction & latency for good users while maximizing fraud capture

🎯 9 EXISTENTIAL BANKING & FIS FRONTIERS (2026–2027)

The Total Space of Fraud: Structural Taxonomy Across All 9 Frontiers

💳 CARD & PAYMENT

- ⏱️ **SLA:** <30ms Inline
- 🔍 **Anomaly:** Intra-seq drift
- 🧠 **Engine:** Transaction FM
- ⚙️ **Tech:** PRAGMA / NeMo
- 🏆 **Target:** Intesa Sanpaolo
- 🎯 **Impact:** Retires XXX+ rules

⚡ INSTANT RAILS

- ⏱️ **SLA:** <50ms Streaming
- 🔍 **Anomaly:** Multi-hop fan-out
- 🧠 **Engine:** Streaming Graph FM
- ⚙️ **Tech:** cuGraph-Stream
- 🏆 **Target:** NPCI (15B+ txs)
- 🎯 **Impact:** Halts smurfing loops

🛡️ ACCOUNT TAKEOVER

- ⏱️ **SLA:** Sub-sec Session
- 🔍 **Anomaly:** Baseline collapse
- 🧠 **Engine:** Hybrid TFM + TabFM
- ⚙️ **Tech:** PyTorch Frame + NeMo
- 🏆 **Target:** Santander & Global
- 🎯 **Impact:** Stops credential runs

🕵️ APP SCAMS & RINGS

- ⏱️ **SLA:** Nearline Lakehouse
- 🔍 **Anomaly:** Multi-table collusion
- 🧠 **Engine:** Relational FM
- ⚙️ **Tech:** KumoRFM-2 (PQL)
- 🏆 **Target:** US Tier-1 Banks
- 🎯 **Impact:** Breaks mule clusters

👤 SYNTHETIC ID, LIVENESS & KYC

- ⏱️ **SLA:** Nearline Onboarding
- 🔍 **Anomaly:** Deepfake & PII forgery
- 🧠 **Engine:** Multi-Modal Agent + Graph
- ⚙️ **Tech:** Vision-Nemotron + GNN
- 🏆 **Target:** Global Digital Banks
- 🎯 **Impact:** Halts fabricated accounts

🌐 CROSS-BORDER AML & FRAML

- ⏱️ **SLA:** Batch & Streaming
- 🔍 **Anomaly:** Shell & UBO concealment
- 🧠 **Engine:** UBO Relational Graph FM
- ⚙️ **Tech:** KumoRFM-2 + DiskGraph
- 🏆 **Target:** Commercial Banks
- 🎯 **Impact:** Traces circular flows

🏢 MERCHANT BUST-OUT

- ⏱️ **SLA:** Real-Time & Daily Batch
- 🔍 **Anomaly:** Credit stacking loops
- 🧠 **Engine:** Bipartite Relational FM
- ⚙️ **Tech:** KumoRFM-2 + cuDF
- 🏆 **Target:** Acquirers & Networks
- 🎯 **Impact:** Stops merchant collusions

⚖️ MODEL GOVERNANCE

- ⏱️ **SLA:** Continuous Validation
- 🔍 **Anomaly:** Future-target leakage
- 🧠 **Engine:** Auditable Relational FM
- ⚙️ **Tech:** ANCHOR_TIMESTAMP
- 🏆 **Target:** Federal Reserve (SR 11-7)
- 🎯 **Impact:** 100% audit compliance

🤖 MULTI-AGENT SWARMS

- ⏱️ **SLA:** Dynamic Adaptive
- 🔍 **Anomaly:** Machine-speed attacks
- 🧠 **Engine:** Autonomous Defense Fleet
- ⚙️ **Tech:** Kumo MCP Agent Swarms
- 🏆 **Target:** Tier-1 SOC / Risk Ops
- 🎯 **Impact:** Intercepts AI adversaries

⚡ NVIDIA + KUMO.AI | BLUEPRINT

Production Stack, 5 Primitives & Enterprise Agent Fleets

📦 WHAT WE ALREADY HAVE TODAY

- 🚀 **Kumo 3 & NIMs:** Containerized CPU/GPU microservices on NGC (`nvcr.io/.../kumo-3`)
- 🧠 **KumoRFM-2 & Tabular:** Zero-shot relational FM + PyTorch Frame (999 quantiles)
- 🛡️ **NVIDIA Fraud Blueprint:** GraphSAGE + cuDF ETL + XGBoost + Dynamo-Triton
- ⚡ **CUDA-X Infra:** cuGraph-PyG, WholeGraph memory tiering, DiskGraph NVMe scratch
- 🤖 **Agentic MCP:** `kumo-rfm-mcp` gateway enabling autonomous PQL execution

🔄 THE PIPELINE REVOLUTION & 5 LAYERS

- 🛑 **Legacy Pipeline:** Raw DB ➔ XXX-line SQL ➔ Static Feature Store (80% tax)
- ⚠️ **Revolut SOTA:** Event Stream ➔ 1D Transformer (Blind to multi-hop graphs)
- 🚀 **RTFM Target:** Streaming Rails + Lakehouse ➔ Dynamic Framing ➔ Hybrid RFM. Fuses sequence encoders + relational graph transformers.
- 🏗️ **5-Layer Stack:** L1 (Compute/TEEs) ➔ L2 (Data Fabric) ➔ L3 (RTFM) ➔ L4 (Serving) ➔ L5 (Agents)
- ⏱️ **Production Win:** Sub-30ms inline authorization + Zero data egress

🔍 WHAT WE SHOULD BE BUILDING: 5 PRIMITIVES

- **P1: Sub-15ms Dynamic Subgraph Framing** ➔ cuGraph-Stream local path extraction. TensorRT-GNN & NVLink Cache: Multi-GPU memory pooling for instant rail clearance
- **P2: WholeGraph Tiered Memory** ➔ HBM/RAM/NVMe scaling to 100B+ edges for NPCI. Path-Centric Flow & Directed Motifs: Traces 5-to-10 hop mule chains without GNN over-smoothing
- **P3: Multimodal Schema-Agnostic Tokenizer** ➔ PyTorch Frame + NeMo with encodings for voice, video, tabular, time-series & graphs.
- **P4: Inductive Subgraph Inference Kernel** ➔ Zero-shot scoring on Triton (<30ms)
- **P5: Temporal Edge Pruning Operator & Temporal Positive-Unlabeled Learning** ➔ Enforces ANCHOR_TIMESTAMP (100% SR 11-7)

🤖 THE AUTONOMY ENGINE: 3 AGENT FLEETS

- 💻 **Coding & SWE Agents:**
  - Auto-schema mapping, synthetic tables, and CI/CD test suites
  - 14-day client onboarding; eliminates manual database ETL maintenance
- 🛡️ **Risk Ops & Investigation Agents:** Slashes repetitive manual triage volume while retaining HITL auditability. **Measurement out-of the-box.**: ⚡ Real-Time Interceptor Agent (<10ms), 🕵️ Forensic Investigation Agent, 📜 Regulatory Filing Agent, 🎯 Adversarial Red-Teaming Agent. All with Autonomous GNN-SHAP subgraph attribution & explainability.
- 👤 **CX & Friction Reduction Agents:**
  - Multi-modal fusion (Video IDV, doc OCR, graphs) for seamless green-path routing
  - Eliminates friction and latency for 98% of verified good customers